



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
REVIEW TEST
15-05-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS]

Time : 3 Hours

PAPER-2

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. The Poynting vector $\vec{S}$ of an electromagnetic wave is given by

$$\vec{S} = 942 \text{ W/m}^2 \cos^2 (10x - 3 \times 10^9 t) \hat{i}$$

where x is in meters and t is in seconds.

(A) The electric field can be approx $600 \text{ N/C} \cos(10x - 3 \times 10^9 t) \hat{j}$

(B) The magnetic field can be approx $2 \times 10^{-6} \text{ T} \cos(10x - 3 \times 10^9 t) \hat{i}$

(C) The wave is propagating in positive x direction

(D) The wave is propagating in negative x direction

Space for Rough Work

2. Laser light falls normally on three evenly spaced, very narrow slits. When one of the side slits is covered, the first-order maximum is at 0.60° from the normal. If the center slit is covered and the other two are open,
- (A) the angle of the first-order maximum is approx at 0.30°
 - (B) the order number of the maximum that now occurs at the same angle as the fourth-order maximum did before is 2.
 - (C) Central maxima is in front of center slit.
 - (D) First minima is approximately at 0.15° .
3. As the width of a slit producing a single-slit diffraction pattern is slowly and steadily reduced, how will the diffraction pattern change?
- (A) The width of the central maxima will increase
 - (B) The width of the central maxima will decrease
 - (C) The maximum intensity will increase
 - (D) The maximum intensity will decrease

Space for Rough Work

4. A plane electromagnetic wave with

$$\vec{E} = \frac{1000}{\sqrt{8.85}} \text{ N/C} \sin\left(\frac{2\pi}{5} \times 10^7 x - \frac{6\pi}{5} \times 10^{15} t\right)$$

Where x & t are in SI units, is incident on a metal plate with work function 2.2 eV and photoelectric efficiency 1 %. $hc = 1240 \text{ eV nm}$, $\text{Area} = 2.48 \text{ m}^2$

- (A) The maximum kinetic energy of electrons is 0.28 eV
 (B) The saturation photocurrent is 1.5 A
 (C) There can't be any photoelectrons
 (D) The saturation photocurrent is 3A
5. The reflected light from an oil film floating on water shows destructive interference for light of wavelengths 560 nm, 630 nm, and 700 nm incident along the normal. The index of refraction of the oil is 1.40. What is film's possible thickness? $\mu_{\text{water}} = 1.33$
- (A) 450 nm (B) 3.6 μm (C) 4.5 μm (D) 9 μm
6. The De-Broglie wavelength of an electron moving in a resistor connected with a voltage source with drift speed is λ . Which of the following actions would increase this wavelength λ ? In each of the options, assume that only that parameter is changed without changing other parameters.
- (A) Temperature of the resistor is increased
 (B) A battery of higher potential difference is maintained across the resistor
 (C) We use a wire of longer length
 (D) We use a wire of larger area.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. Assume that a universe exists where plank's constant h , mass of electron, permittivity of free space, electronic charge may be different from our universe. Predict the effect of each of the constant on parameters of hydrogen atom.

List-I		List-II	
(P)	h is more	(1)	Radius of atom is more
(Q)	m_e is lesser	(2)	Radius of atom is less
(R)	ϵ_0 is lesser	(3)	Wavelength of first line of lyman series is larger.
(S)	e is more	(4)	Speed of electron in 1 st orbit is more.
		(5)	Angular momentum in ground state is more

(A) $P \rightarrow 1,3,5; Q \rightarrow 1,5; R \rightarrow 2,4; S \rightarrow 2,4$

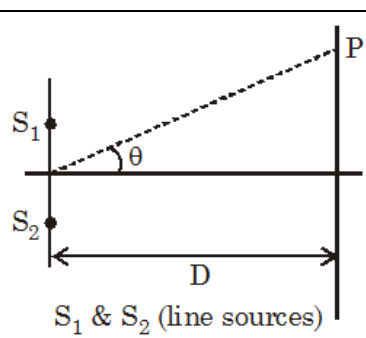
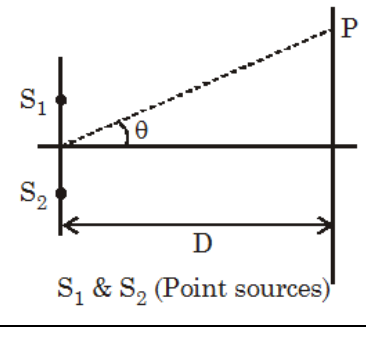
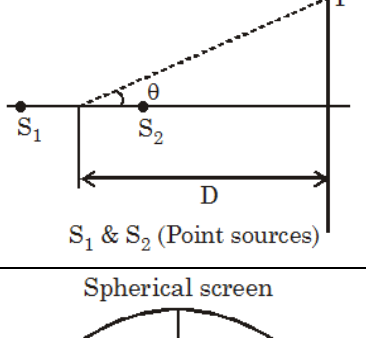
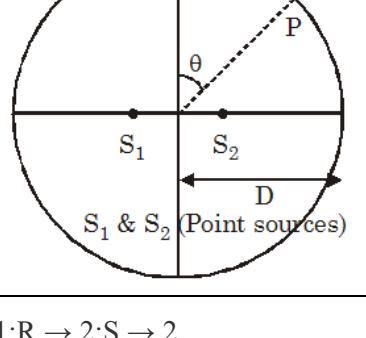
(B) $P \rightarrow 1,3; Q \rightarrow 1,3; R \rightarrow 2,4; S \rightarrow 2,4$

(C) $P \rightarrow 1,3,5; Q \rightarrow 1,3; R \rightarrow 2,3,4; S \rightarrow 2,3,4$

(D) $P \rightarrow 1,3,5; Q \rightarrow 1,3; R \rightarrow 2,4; S \rightarrow 2,4$

Space for Rough Work

8.

List-I		List-II (Fringes shape on screen)	
(P)	 S_1 & S_2 (line sources)	(1)	Hyperbolic
(Q)	 S_1 & S_2 (Point sources)	(2)	Circular
(R)	 S_1 & S_2 (Point sources)	(3)	Straight line
(S)	 Spherical screen S_1 & S_2 (Point sources)	(4)	Parabolic

(A) $P \rightarrow 1; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 2$

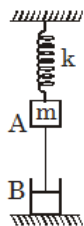
(B) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 2$

(C) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 2; S \rightarrow 1$

(D) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 2; S \rightarrow 2$

Space for Rough Work

9. In the figure shown, spring has an initial extension x_0 , dashpot exerts a force $F = -Cv$, where v represent velocity. Dashpot is fixed to the ground. Except mass of block, all devices have negligible mass. Comment on the nature of vertical motion for different values of x_0 , k , m , C . The block is released from rest.



List-I		List-II	
(P)	AB is a rod, $x_0 = 0$, $k = 100$ N/m, $m = 1$ kg, $C = 50$ kg/sec.	(1)	undamped SHM
(Q)	AB is a string, $x_0 = 0$, $k = 100$ N/m, $m = 1$ kg, $C = 50$ kg/sec.	(2)	damped harmonic motion.
(R)	AB is a string, $x_0 = 10$ cm, $k = 100$ N/m, $m = 1$ kg, $C = 50$ kg/sec.	(3)	critically damped motion.
(S)	AB is a rod, $x_0 = 5$ cm, $k = 25$ N/m, $m = 1$ kg, $C = 10$ kg/sec.	(4)	overdamped motion.
		(5)	no motion.

(A) $P \rightarrow 4; Q \rightarrow 4; R \rightarrow 5; S \rightarrow 2$

(B) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 5; S \rightarrow 1$

(C) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 5; S \rightarrow 3$

(D) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$

Space for Rough Work

10. In List-I, we have listed certain quantities for an electron in Bohr atom. Match their dependence on n & Z in list-II.

List-I		List-II	
(P)	magnetic dipole moment	(1)	Z^0
(Q)	magnetic field at center of circular path	(2)	Z^2
(R)	angular velocity	(3)	Z^3
(S)	potential energy (assuming standard reference point)	(4)	n^1
		(5)	n^{-3}
		(6)	n^{-2}

- (A) $P \rightarrow 1,4; Q \rightarrow 3,5; R \rightarrow 2,5; S \rightarrow 2,6$ (B) $P \rightarrow 1,4; Q \rightarrow 3; R \rightarrow 2,5; S \rightarrow 2,6$
 (C) $P \rightarrow 1,6; Q \rightarrow 3; R \rightarrow 2,5; S \rightarrow 2,6$ (D) $P \rightarrow 1,4; Q \rightarrow 3; R \rightarrow 3,5; S \rightarrow 2,5$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

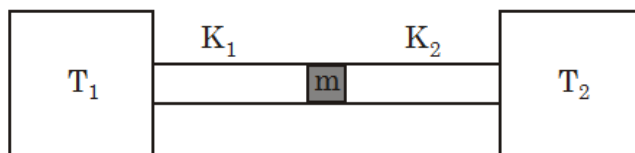
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1. An electron is projected at an angle of 53° to a uniform electric field of 9 n N/C . Take $m_e = 9 \times 10^{-31} \text{ kg}$. What will be its de Broglie wavelength (in nm) after 1 sec if its de Broglie wavelength initially is 220 nm. Take : $h = 6.6 \times 10^{-34} \text{ kgm}^2/\text{sec}^2$.
 2. Ten moles of a gas (molar heat capacity for constant volume process is C_V) is enclosed in rigid hollow sphere of inner radius 'a' and outer radius '5a' and its temperature is $3T_0$ at $t = 0$. Heat is conducted out to the environment (temperature T_0) through the sphere material of conductivity K and negligible heat capacity. At $t = t_1$ (second) the temperature of the gas is found to be $2T_0$ then find the value of t_1 . [All quantities are expressed in SI units and take $C_V \times (\ell n_2) = 2\pi Ka$]
-

Space for Rough Work

3. An astronomer observed with his telescope two stars, A and B. Star A has a spectral peak at $3500 \text{ \AA}$ and star B has a spectral peak at $7000 \text{ \AA}$. If they appear equally bright (emitting equal energy per unit time) what is the ratio of radius of the star B to that of A.
4. In an experiment on photoelectric effect, the slope of the cut-off voltage (y-axis) versus frequency (x-axis) of incident light is found to be $4.12 \times 10^{-15} \text{ V s}$. Calculate the value of Planck's constant (in 10^{-15} eV-s).
5. A thermostatted chamber at small height h above earth's surface maintained at 30°C has a clock fitted in it with an uncompensated pendulum. The clock designer correctly designs it for height h , but for temperature of 20°C . If this chamber is taken to earth's surface, the clock in it would click correct time. Find the coefficient of linear expansion of material of pendulum. (earth's radius is R). If the coefficient is h/nR , find the value of n .

Space for Rough Work

6. A wire of length 1.0m and radius 10^{-3} m is carrying a heavy current and is assumed to radiate as a black body. At equilibrium its temperature is 900K while that of the surroundings is 300K. The resistivity of the material of the at 300K is $\pi^2 \times 10^{-8}$ ohm-m and its temperature coefficient of resistance is $7.8 \times 10^{-3}/^\circ\text{C}$. Find the current (in A) in the wire. [Given: Stefan's constant = 5.68×10^{-8} W/m²K⁴]
7. A new monatomic ideal gas is discovered and named "Wellsium". A pure 4-mole sample is sitting in a container at equilibrium in a 20.0°C environment. According to the kinetic theory of gases, the average kinetic energy per molecule for this gas is $Z \times 10^{-21}$ J. The value of $Z/3$ to the nearest integer is : (Boltzman's constant is 1.38×10^{-23} J/K)
8. A block of mass m (having high conductivity) and specific heat capacity S is connected to constant temperature sources at temperature $T_1 = 4 T_0$ and $T_2 = 2T_0$ respectively by conducting rods of negligible heat capacity. The length and area of cross-section of rods are L and A respectively. The conductivity of rods are $K_1 = K$ & $K_2 = 2K$ as shown in fig. Neglecting any radiation loss, find the temperature of block (in K) at time $\frac{mSL\ln 2}{KA}$. Temperature of block at $t = 0$ is T_0 . Take : $T_0 = 480$ K



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

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Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Element "X" forms a compound $\text{XCl}_2 \cdot 4\text{NH}_3$ that does not show any type of isomerism. Element "Y" forms $\text{YCl}_3 \cdot 4\text{NH}_3$ type compound which has two isomers "P" & "Q". Also, both elements "X" & "Y" belong to VIII group of Mendeleev's periodic table.

"Y" is the only metal present in vitamin B_{12} and compounds of "X" inhibit growth of tumours.

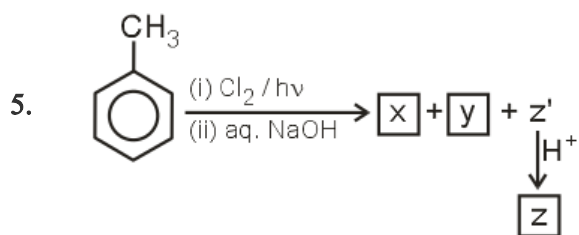
Which is/are correct about X & Y or their compounds?

- (A) If one NH_3 is removed from the formula of $\text{YCl}_3 \cdot 4\text{NH}_3$, then one of the isomer of newly formed complex compound becomes optically active
- (B) Both "P" & "Q" are 1 : 1 electrolytes
- (C) If two NH_3 in $\text{XCl}_2 \cdot 4\text{NH}_3$ get replaced by Cl then the formula of the new compound will show stereoisomerism
- (D) X and X^{2+} have same number of unpaired electrons

Space for Rough Work

2. Which of the following is/are correct?
- (A) AsH_3 is better lewis base than NH_3
- (B) $(d_{\text{N-Cl}})_{\text{NOCl}_3} > (d_{\text{N-F}})_{\text{NOF}_3}$
- (C) B_2H_6 on reaction with CH_3OCH_3 will maintain the same hybridisation of Boron in the final product
- (D) There is no hybridisation in SbH_3 hence the molecule does not exist
3. For the process : $\text{H}_2\text{O}(\text{s}, -10^\circ\text{C}, 1\text{atm}) \rightarrow \text{H}_2\text{O}(\text{g}, 473^\circ\text{C}, 1\text{atm})$, the correct information(s) is/are -
- Given : $C_{p,m}$ values (cal/K-mole) $\text{H}_2\text{O}(\text{s}) = 9.0$, $\text{H}_2\text{O}(\text{l}) = 18.0$, $\text{H}_2\text{O}(\text{g}) = 8.0$
 Latent heat of fusion of ice at $0^\circ\text{C} = 1365$ cal/mole
 Latent heat of vapourisation of water at $100^\circ\text{C} = 9698$ cal/mol
 $\ln\left(\frac{273}{263}\right) = 0.04$, $\ln\left(\frac{373}{273}\right) = 0.30$, $\ln\left(\frac{746}{373}\right) = 0.70$.
- (A) $\Delta H = 15937$ cal/mole (B) $\Delta H = 16310$ cal/mole
 (C) $\Delta S = 42.36$ cal/K-mol (D) $\Delta S = 2.36$ cal/K-mol
4. Identify which of the following solution(s) will show negative deviation from Raoult's law -
- (A) Carbon-disulphide + acetic acid (B) Water + Formic Acid
 (C) Propan-2-ol + 2-methyl propanol (D) Chloroform + Acetone

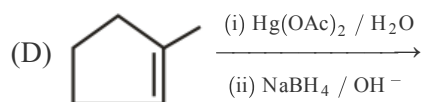
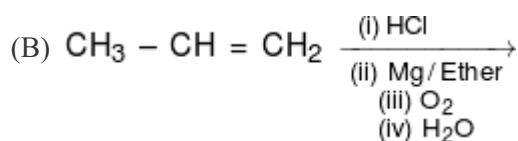
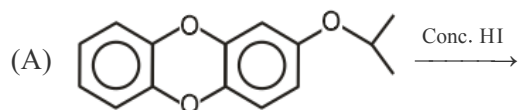
Space for Rough Work



(B.P.) $z > y > x$

Select correct option(s)

- (A) "z" evolved CO_2 gas with NaHCO_3 .
 (B) "y" gives immediate turbidity with Lucas reagent.
 (C) "x" change colour of $\text{Br}_2 + \text{H}_2\text{O}$
 (D) All compound (x, y and z) give H_2 gas with Na metal.
6. In which reaction(s) alcohol can formed as one of the product



Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. Match the List.

List-I Complex Compound/Ion		List-II Characteristics	
(P)	$[\text{Zn}(\text{gly})_2]$	(I)	Exhibits Geometrical Isomerism only
(Q)	$[\text{Pd}(\text{NH}_2\text{CH}(\text{CH}_3)\text{COO}^-)_2]$	(II)	Exhibits Optical isomerism only
(R)	$[\text{Co}(\text{C}_2\text{S}_2\text{O}_2)_2(\text{NH}_3)_2]^-$	(III)	Both geometrical & optical isomerism are exhibited
(S)	$[\text{Cu}(\text{NH}_3)_4][\text{CuCl}_4]$	(IV)	Structural isomerism is exhibited
		(V)	Square planar around any one central atom
		(VI)	Tetrahedral around any one central atom

Which of the following option is correct match?

(A) $\text{P} \rightarrow \text{VI}; \text{Q} \rightarrow \text{III}; \text{R} \rightarrow \text{III}; \text{S} \rightarrow \text{VI}$

(B) $\text{P} \rightarrow \text{IV}; \text{Q} \rightarrow \text{V}; \text{R} \rightarrow \text{IV}; \text{S} \rightarrow \text{V}$

(C) $\text{P} \rightarrow \text{V}; \text{Q} \rightarrow \text{II}; \text{R} \rightarrow \text{VI}; \text{S} \rightarrow \text{I}$

(D) $\text{P} \rightarrow \text{II}; \text{Q} \rightarrow \text{I}; \text{R} \rightarrow \text{I}; \text{S} \rightarrow \text{III}$

Space for Rough Work

8. Match the List.

List-I		List-II	
(P)	NaCl structure	(1)	Cation forming fcc and anion at octahedral void
(Q)	Zinc Blende structure	(2)	Anion forming fcc and at all octahedral void while cation goes to all tetrahedral void
(R)	Cesium Halide structure	(3)	Anion forming fcc and cation at alternate tetrahedral void
(S)	Flourite structure	(4)	Cation forming fcc and anion at alternate tetrahedral void.
		(5)	Anion forming simple cube & cation at cubical void.
		(6)	Cation forming fcc & anion at all tetrahedral void.

Which of the following option is correct match?

 (A) $P \rightarrow 1; Q \rightarrow 3, 4; R \rightarrow 2, 5; S \rightarrow 6$

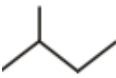
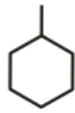
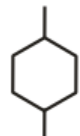
 (B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 6$

 (C) $P \rightarrow 3, 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 6$

 (D) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2, 5; S \rightarrow 1$

Space for Rough Work

9.

Column-I Compound		Column-II Number of mon-chlorinated product(s) (only structural)	
(P)		(I)	2
(Q)		(II)	3
(R)		(III)	4
(S)		(IV)	5

(A) $P \rightarrow \text{III}; Q \rightarrow \text{I}; R \rightarrow \text{IV}; S \rightarrow \text{II}$

(B) $P \rightarrow \text{II}; Q \rightarrow \text{III}; R \rightarrow \text{I}; S \rightarrow \text{IV}$

(C) $P \rightarrow \text{I}; Q \rightarrow \text{III}; R \rightarrow \text{IV}; S \rightarrow \text{II}$

(D) $P \rightarrow \text{IV}; Q \rightarrow \text{III}; R \rightarrow \text{II}; S \rightarrow \text{I}$

10. For the reaction

$2A(g) \rightarrow B(g)$, starting with 4 moles of A & half life 40 sec in a 5 litre container at 400 K.

	List-I		List-II
(P)	If reaction follow first order then initial rate of reaction	(1)	$3.46 \times 10^{-4} \text{ M/s}$
(Q)	If reaction follow zero order then rate of reaction at time : $t = 40 \text{ sec}$.	(2)	$6.93 \times 10^{-3} \text{ M/s}$
(R)	If reaction follow first order kinetics then rate of reaction at 95% completion.	(3)	$5 \times 10^{-3} \text{ M/s}$
(S)	If reaction follow second order kinetics then rate of reaction at 99% completion	(4)	10^{-6} M/s
		(5)	$6.25 \times 10^{-5} \text{ M/s}$

(A) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$

(B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

(C) $P \rightarrow 5; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$

(D) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Number of reactions in which atleast one gaseous product is formed.
 - (i) $2\text{CaO} \cdot \text{MgO} + \text{Fe-Si} \xrightarrow{1150^\circ\text{C}}$
 - (ii) $4\text{Au} + 8\text{NaCN} + \text{O}_2 + 2\text{H}_2\text{O} \rightleftharpoons$
 - (iii) $\text{SnSiO}_3 + \text{CaO} + \text{C} \rightarrow$
 - (iv) $2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow$
 - (v) $\text{XeF}_2 + \text{H}_2\text{O} \rightarrow$
 - (vi) $\text{ZnO} + \text{C (coke)} \xrightarrow{1673\text{K}}$
2. During extraction of Magnesium by Dow's process, magnesium hydroxide is treated with HCl followed by concentration of solution and crystallisation to give crystals of $\text{MgCl}_2 \cdot x\text{H}_2\text{O}$. The value of "x" is _____.
3. Number of specie(s) that use atleast one double dumbell shape orbital for hybridization of central atom.

(i) $[\text{PtCl}_4]^{2-}$	(ii) $\text{Ni}(\text{CO})_4$
(iii) PCl_5	(iv) $[\text{CoF}_6]^{3-}$
(v) $\text{K}_4[\text{Ni}(\text{CN})_4]$	(vi) XeF_6
(vii) $\text{Fe}(\text{CO})_5$	(viii) SO_2Cl_2

Space for Rough Work

4. A solid cube with edge length of 2.532 cm of an ionic compound which has NaCl type lattice is added to 1kg of water. The boiling point of this solution is found to be 100.52°C (Assume 100% ionisation of ionic compound). If radius of anion of ionic solid is 2Å, then calculate radius of cation in Å.

Given : K_b of water = 0.52 K·kg mol⁻¹ ,

$$N_A = 6 \times 10^{23}, \sqrt[3]{216} = 6, (2.532)^3 \times \frac{4}{3} = 21.6$$

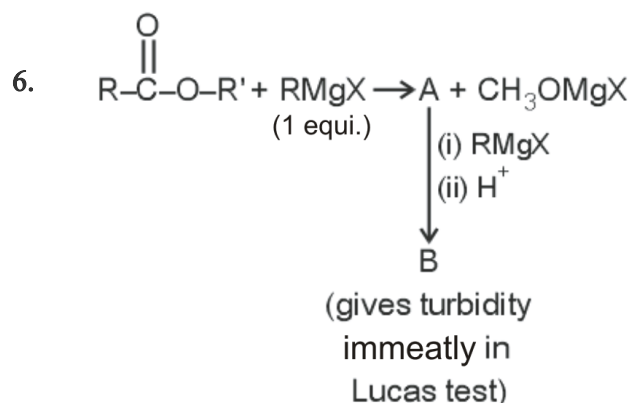
5. $\Lambda_{eq}^0 \text{ BaCl}_2 = 160 \text{ cm}^2 \text{ eq}^{-1} \text{ ohm}^{-1}$

$$\Lambda_{eq}^0 \text{ K}_3\text{PO}_4 = 140 \text{ cm}^2 \text{ eq}^{-1} \text{ ohm}^{-1}$$

$$\Lambda_{eq}^0 \text{ KCl} = 100 \text{ cm}^2 \text{ eq}^{-1} \text{ ohm}^{-1}$$

$$K_{sp} \text{ of Ba}_3(\text{PO}_4)_2 = X \times 10^{-25} \text{ M}^5$$

$\kappa = 12 \times 10^{-6} \text{ ohm}^{-1} \text{ cm}^{-1}$ for $\text{Ba}_3(\text{PO}_4)_2$ solution. Give 'X'.



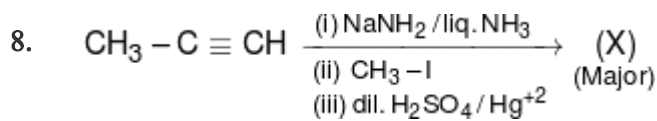
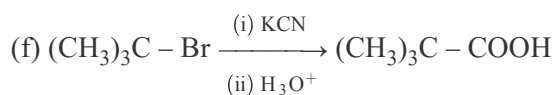
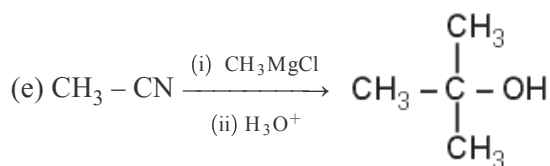
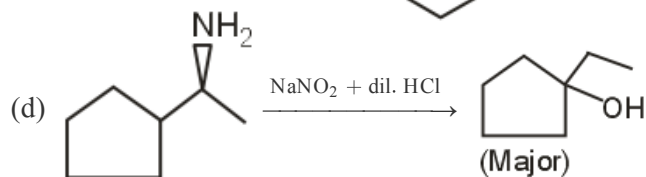
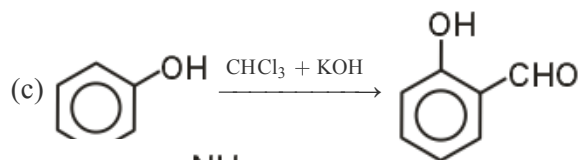
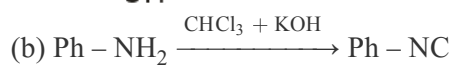
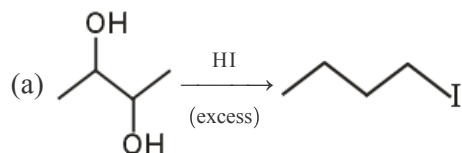
If p = molecular mass of group $\boxed{\text{R}}$

q = molecular mass of group $\boxed{\text{R}'}$

where $\boxed{\text{R}}$ and $\boxed{\text{R}'}$ are alkyl group's then minimum value of $|p-q|$ is

Space for Rough Work

7. How many reaction(s) are correct?



Total D-exchange, when (X) treated with $\text{OD}^- / \text{D}_2\text{O}$.

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Let $I = \int \frac{dx}{a + b \cos x}$ where $a, b > 0$ and $a + b = \lambda_1$ and $a - b = \lambda_2$ then
(where c is the constant of integration)

(A) If $\lambda_2 = 0$ then $I = \frac{2}{\lambda_1} \tan \frac{x}{2} + c$

(B) If $\lambda_2 > 0$ then $I = \frac{1}{\sqrt{\lambda_1 \lambda_2}} \tan^{-1} \left(\sqrt{\frac{\lambda_2}{\lambda_1}} \tan \frac{x}{2} \right) + c$

(C) If $\lambda_2 < 0$ then $I = \frac{2}{\sqrt{-\lambda_1 \lambda_2}} \log_e \left| \frac{\sqrt{\lambda_1} + \sqrt{-\lambda_2} \tan \frac{x}{2}}{\sqrt{\lambda_1} - \sqrt{-\lambda_2} \tan \frac{x}{2}} \right| + c$

(D) If $\lambda_2 < 0$ then $I = \frac{1}{\sqrt{-\lambda_1 \lambda_2}} \log_e \left| \frac{\sqrt{\lambda_1} + \sqrt{-\lambda_2} \tan \frac{x}{2}}{\sqrt{\lambda_1} - \sqrt{-\lambda_2} \tan \frac{x}{2}} \right| + c$

Space for Rough Work

2. Let $f(x)$ be a cubic polynomial which is having local maximum at $(1, 2)$ and $f'(x)$ has local extremum at $x = 0$. If $f(0) = 1$ then
- (A) Number of real roots of $f(x) - f(-x) = 6x - 10$ is 1
- (B) If $g(x) + f(x) = 0$ then complete interval in which $g(x)$ is increasing is $x \in (-\infty, -1] \cup [1, \infty)$
- (C) The range of values of K for which $f(x) = K$ has three distinct solutions is $K \in (0, 2)$
- (D) Range of the function $P(\theta) = f(\sin\theta)$ where $\theta \in \mathbb{R}$ is $[0, 2]$
3. For the curve $\sin x + \sin y = 1$, $\lim_{x \rightarrow 0} (x^\alpha) \frac{d^2y}{dx^2}$ exists then α can be
- (A) $\frac{3}{2}$ (B) 2
- (C) $\frac{5}{2}$ (D) 1

Space for Rough Work

4. If $f(x) = \lim_{n \rightarrow \infty} \frac{2x^{2n} + x + 2}{x^{2n} - x^2 + 3}$, then which of the following statement(s) holds good ?
- (A) $f(x)$ is continuous at $x = 1$ (B) $f(x)$ is discontinuous at $x = 1$
 (C) $\lim_{x \rightarrow 2} f(x)$ exists finitely (D) $\lim_{x \rightarrow 2} f(x)$ does not exist
5. Let $a \in \mathbb{R}$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = 3x^5 - 50x^3 + 135x - a$. The equation $f(x) = 0$ will have :
- (A) one real root for $|a| > 216$ (B) three real roots for $88 < |a| < 216$
 (C) five real roots for $|a| < 88$ (D) five real roots for $a \in \mathbb{R}$
6. If $\int \frac{x^6 - x^3 - 2}{x^3 \sqrt{(x^3 + 1)^2 + 3x^4}} dx = f(x) + C$, then -
- (A) $f(1) < f(-1)$ (B) $f(2) > f(3)$ (C) $f(1) > f(-1)$ (D) $f(2) < f(3)$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each **question has matching lists**. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

7. Let $\phi(x) = 3f\left(\frac{x^2}{3}\right) + f(3 - x^2) \forall x \in (-3, 4)$ where $f''(x) > 0 \forall x \in (-3, 4)$,

$$g(x) = (a^2 - 3a + 2) \left(\cos^2 \frac{x}{4} - \sin^2 \frac{x}{4} \right) + (a - 1)x + \sin 1$$

Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	$\phi(x)$ is increasing in	(1)	$(0, 1) \cup (1, 4)$
(Q)	$\phi(x)$ is decreasing in	(2)	$\left[-\frac{3}{2}, 0\right] \cup \left[\frac{3}{2}, 4\right)$
(R)	The set of values of a for which $g(x)$ does not possess critical point is	(3)	$\left(-3, -\frac{3}{2}\right] \cup \left[0, \frac{3}{2}\right]$
(S)	The set of values of a for which $g(x)$ possess critical point is	(4)	$(-\infty, 0] \cup \{1\} \cup [4, \infty)$

(A) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$

(B) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$

(C) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 4$

(D) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$

Space for Rough Work

8. Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	If $x = \alpha$ satisfies $\sin^{-1}x + \cos^{-1}x^2 + \frac{\pi}{2} = 0$, then the value of $\frac{\sec^{-1}\alpha - \tan^{-1}\alpha}{\cot^{-1}\alpha - \operatorname{cosec}^{-1}\alpha}$ is	(1)	0
(Q)	If $a = \lim_{x \rightarrow \infty} (2.3^x + 3.4^x + 4.5^x)^{\frac{1}{x}}$ and $b = \lim_{x \rightarrow (-\infty)} (2.3^x + 3.4^x + 4.5^x)^{\frac{1}{x}}$ then $\left(\frac{a+b}{a-b}\right)$ is equal to	(2)	4
(R)	Number of points of discontinuity of $f(x) = x \sin(\pi(2x - [x]))$ in $[0, 2]$ will be (Note $[k]$ denotes greatest integer function less than or equal to k)	(3)	1
(S)	$\lim_{x \rightarrow 0} \left(\frac{3e^{x/3} - x - 3}{x^2} \right)$ is equal to	(4)	$\frac{1}{3}$
		(5)	$\frac{1}{2}$
		(6)	$\frac{1}{6}$

(A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 2; S \rightarrow 4$

(B) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 6$

(C) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 6$

(D) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$

Space for Rough Work

9. Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	If $g(x) = f(-x + xf(x))$ and $f(0) = 1, f'(0) = 2$, then value of $g'(0)$ is equal to	(1)	$\frac{1}{4}$
(Q)	If $e^y = e^x x^{-y}$, then $\left(\frac{dy}{dx}\right)_{x=e}$ is equal to	(2)	6
(R)	Number of points of discontinuity of $y = [2x]$ in $[-1, 2]$ is equal to (where $[.]$ denote greatest integer function)	(3)	0
(S)	If $I = \lim_{x \rightarrow 0} (1 + \tan x)^{\frac{1}{\sin x}}$ then $[I]$ is equal to (where $[.]$ denote greatest integer function)	(4)	2
		(5)	3

(A) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$ (B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 5$

(C) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 5$ (D) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 5$

10. If a curve $f(x, y, a, b) = 0$ is given by $\sin^{-1}x + \cos^{-1}ay + \cos^{-1}(bxy) = \frac{\pi}{2}$

Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	if $a = 1, b = 2$, then	(1)	$(1 - x^2)(1 - y^2) = x^2y^2$
(Q)	if $a = 2, b = 1$, then	(2)	$a + b = 8$
(R)	if $a = 0, b = 1$, then	(3)	$\frac{dy}{dx} = -\left(\frac{y^2 + 1}{xy}\right), x, y \in \mathbb{R}_0$
(S)	if $a = b$ and area bounded by rectangle formed by $f(x, y, a, b) = 0$ is 1 sq. unit, then	(4)	$x^2 + 4y^2 = 1 + 3x^2y^2$

(A) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$ (B) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 2$

(C) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 2$ (D) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 1$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Let $\int \frac{(x^2 - 1) dx}{x^3 \sqrt{3x^4 + 2x^2 - 1}} = f(x) + c$, (where c is constant of integration) and $f(1) = -1$. If $\lambda = \lim_{x \rightarrow \infty} f(x)$ then the absolute value of $\frac{\lambda}{\sqrt{3}}$
2. If the real root of the equation $(x - 1)^{2019} + (x - 2)^{2019} + (x - 3)^{2019} + \dots + (x - 2019)^{2019} = 0$ is λ , then $\frac{\lambda}{315}$ equals

Space for Rough Work

3. $f(x) = \tan(\sin^{-1}(\cos(\sin^{-1}(x))) \tan(\cos^{-1}(\sin(\cos^{-1}x)))$, where $x \in (0, 1)$ then $\sum_{r=0}^{\infty} \frac{1}{2^r} f\left(\frac{1}{2^r}\right)$ is equal to
4. The value of $\lim_{n \rightarrow \infty} \left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{n^2}\right)$ is
5. Let p_1 and p_2 be two values of p for which $f(x) = \begin{cases} \frac{x(\ln(1+x) + \ln(1-x))}{(\sec x - \cos x)}; & x \in (-1, 0) \\ (p^2 - 3p - 1) \sin x + x^2; & x \in [0, \infty) \end{cases}$ is differentiable at $x = 0$ find $p_1^2 + p_2^2$

Space for Rough Work

6. Let $P(x)$ be a polynomial of degree 4 with $P(1) = 1$, $P'(1) = 0$, $P''(1) = 2$, $P'''(1) = -12$ and $P''''(1) = 24$ then value of $\frac{P(2)}{P(3)}$ is equal to
7. A curve is represented by $x = \sec^2 t$ and $y = \cot t$, where t is the parameter. If the tangent at point P on curve where $t = \frac{\pi}{4}$ meets the curve again at Q then $|PQ| = \frac{p\sqrt{q}}{2}$, where p, q are coprime. Find $p + q$
8. Let f be a real valued function such that $f\left(\frac{2x-3}{x-2}\right) = 5x - 2$, $x \neq 2$, then value of $\frac{5}{4}f^{-1}(13)$ is

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work